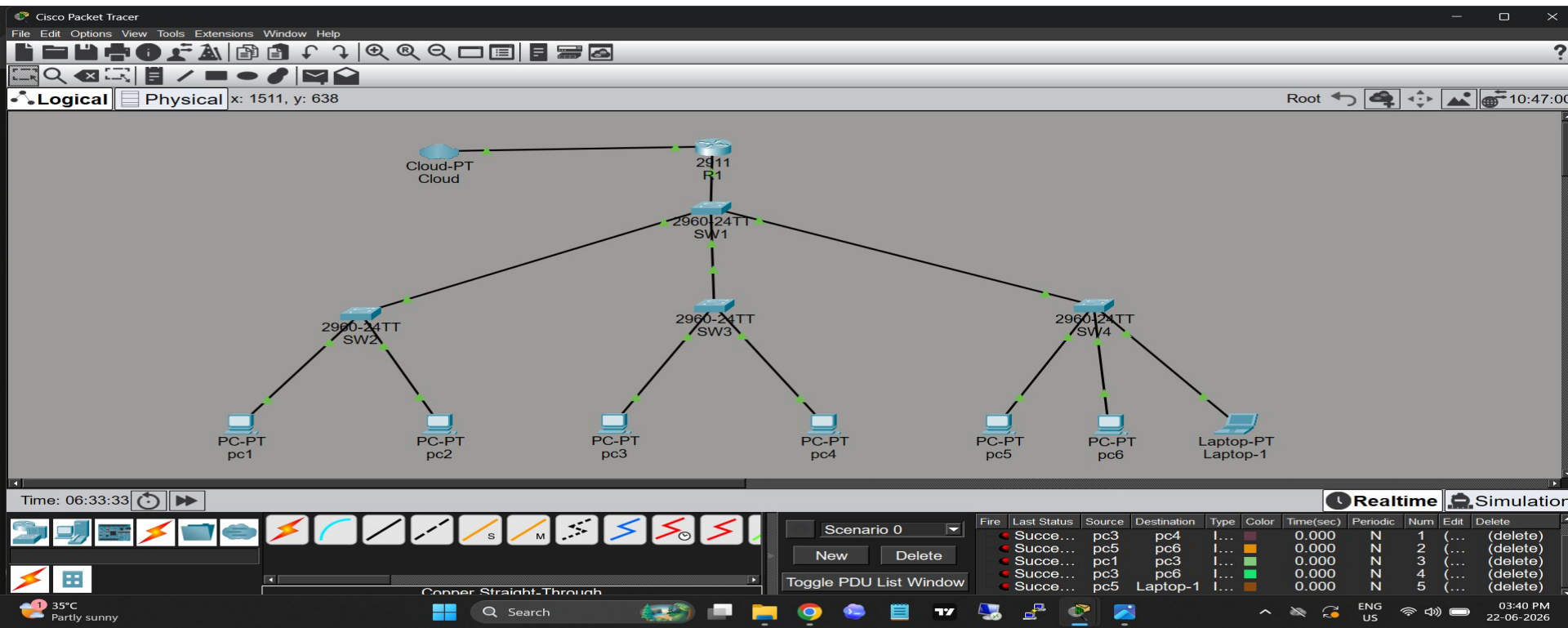


Enterprise Network Design

TOPOLOGY

- ❖ 1 Router (R1 - 2911) + Cloud-PT
- ❖ 1 Core Switch (SW1) + 3 Access Switches (SW2, SW3, SW4)
- ❖ 6 PCs + 1 Laptop connected across access switches



PORT SECURITY

Configured on SW2

- ❖ **Ports: Fa0/1, Fa0/2**
- ❖ **Max MAC addresses: 1**
- ❖ **Violation mode: Shutdown**
- ❖ **Verified with: show port-security**

```
SW2#  
SW2#show port-security  
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action  
          (Count)          (Count)          (Count)  
-----  
      Fa0/1         1         1             0      Shutdown  
      Fa0/2         1         0             0      Shutdown  
-----  
SW2#
```

CONNECTIVITY TEST

- ❖ Ping from PC to 192.168.20.1
- ❖ Result: 4 sent, 4 received, 0% loss
- ❖ Inter-VLAN routing confirmed working

```
C:\>ping 192.168.20.1
```

```
Pinging 192.168.20.1 with 32 bytes of data:
```

```
Reply from 192.168.20.1: bytes=32 time=3ms TTL=255
```

```
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
```

```
Reply from 192.168.20.1: bytes=32 time=1ms TTL=255
```

```
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
```

```
Ping statistics for 192.168.20.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

```
C:\>ssh -l admin 192.168.10.1
```

```
Password:
```

```
R1#
```

REMOTE ACCESS

- ❖ **Connected to R1 remotely**
- ❖ **Command: `ssh -l admin 192.168.10.1`**
- ❖ **Login successful → R1#**

```
C:\>ssh -l admin 192.168.10.1
```

```
Password:
```

```
R1#|
```

DHCP TEST

- ❖ PC requested IP automatically
- ❖ Result: DHCP request successful
- ❖ IP: 192.168.10.2
- ❖ Gateway: 192.168.10.1
- ❖ DNS: 8.8.8.8

Physical Config Desktop Programming Attributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

DHCP

Static

DHCP request successful.

IPv4 Address

192.168.10.2

Subnet Mask

255.255.255.0

Default Gateway

192.168.10.1

DNS Server

8.8.8.8

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::2D0:FFFF:FE1B:DE1C

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

VLANs

VLAN 10 → IT

VLAN 20 → HR

VLAN 30 → SALES

VLAN 40 → GUEST

Verified with: show vlan brief

All VLANs active

```
SW2>enable
SW2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	IT	active	Fa0/1, Fa0/2
20	HR	active	
30	SALES	active	
40	GUEST	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW2#
```

INTER VLAN ROUTING

Method: Router on a Stick

- ❖ Sub-interfaces on R1:
- ❖ Gi0/0.10 → 192.168.10.1
- ❖ Gi0/0.20 → 192.168.20.1
- ❖ Gi0/0.30 → 192.168.30.1
- ❖ Gi0/0.40 → 192.168.40.1

```
R1>enable
R1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	up	up
GigabitEthernet0/0.10	192.168.10.1	YES	manual	up	up
GigabitEthernet0/0.20	192.168.20.1	YES	manual	up	up
GigabitEthernet0/0.30	192.168.30.1	YES	manual	up	up
GigabitEthernet0/0.40	192.168.40.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	up	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```
R1#
```

TRUNKING

- ❖ 4 trunk ports on SW1: Fa0/1, Fa0/3, Fa0/4, Fa0/5
- ❖ Encapsulation: 802.1Q
- ❖ VLANs allowed: 1, 10, 20, 30, 40
- ❖ Verified with: show interfaces trunk

```
SW1>enable
SW1#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/1     on        802.1q         trunking      1
Fa0/3     on        802.1q         trunking      1
Fa0/4     on        802.1q         trunking      1
Fa0/5     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/1     1-1005
Fa0/3     1-1005
Fa0/4     1-1005
Fa0/5     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30,40
Fa0/3     1,10,20,30,40
Fa0/4     1,10,20,30,40
Fa0/5     1,10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30,40
Fa0/3     1,10,20,30,40
Fa0/4     1,10,20,30,40
--More--
```

THANK YOU